Digital Image Processing — Unit 2

Techniques · Segmentation · Clustering · Fuzzy Logic · Neural Networks

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Course Page:

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Course Outline

Part A: Foundations

- Introduction to Digital Images
- Application Areas of DIP
- Binarization of Digital Images
- Global vs. Adaptive Thresholding

Part B: Segmentation

- Image Segmentation Techniques
- Clustering Approaches
- K-Means Algorithm
- Fuzzy C-Means Algorithm

Part C: Advanced Methods

- Fuzzy Logic in DIP
- Neural Networks in DIP
- Problems with Color Images
- Object Extraction from Color Images

Exam Questions Covered

- Questions 7–16 with detailed solutions

Part A

Introduction to Digital Images & Binarization

What is a Digital Image?

Definition

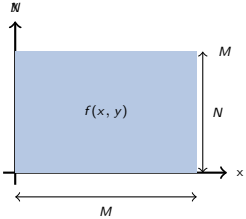
A digital image is a **2D (or 3D) matrix of discrete picture elements (pixels)**, where each pixel stores an intensity or color value.

Key Attributes:

- **Resolution**: Number of pixels ($M \times N$)
- **Bit Depth**: Bits per pixel (8-bit $\rightarrow$ 256 gray levels)
- **Color Model**: Grayscale, RGB, HSV, CMYK
- **Spatial Frequency**: Rate of intensity change

90	105	120	135	150
75	90	105	120	135
60	75	90	105	120
45	60	75	90	105
30	45	60	75	90

Pixel Grid (5×5)



Mathematical Representation

$f(x, y)$ where x, y = spatial coordinates,
f = intensity at (x, y)

For RGB: $\mathbf{f}(x, y) = [R(x, y),\ G(x, y),\ B(x, y)]$

Application Areas of Digital Image Processing

Medical Imaging

- MRI, CT, X-ray enhancement
- Tumor detection & segmentation
- Retinal image analysis
- Histopathology slides

Industrial Inspection

- Product defect detection
- PCB quality control
- Automated measurement
- Robotic vision systems

Remote Sensing

- Satellite image analysis
- Land use/cover classification
- Disaster monitoring
- Crop yield prediction

Security & Surveillance

- Face recognition
- Fingerprint analysis
- License plate recognition
- Crowd monitoring

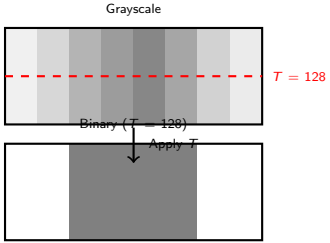
Also: Astronomy, Document Analysis (OCR), Entertainment, Autonomous Vehicles

Binarization of a Digital Image

Concept

Binarization converts a **grayscale image** into a **binary image** (only 0 or 255) by applying a threshold T:

$$B(x, y) = \begin{cases} 255 & \text{if } f(x, y) \geq T \\ 0 & \text{if } f(x, y) < T \end{cases}$$



Why Binarization?

- Simplifies image data for analysis
- Reduces computational complexity
- Essential for OCR, shape analysis
- Separates foreground from background
- Enables morphological operations

Key Challenge

Selecting the optimal threshold T — too high misses foreground, too low includes background noise.

Binarization Methods

1. Global Thresholding

Single threshold for **entire image**.

Otsu's Method: Maximizes inter-class variance:

$$\sigma_B^2 = \omega_0\omega_1(\mu_0 - \mu_1)^2$$

Pros: Simple, fast
Cons: Fails with uneven illumination

2. Local/Adaptive Thresholding

Threshold varies per **local region**:

$$T(x, y) = \mu_{local}(x, y) - C$$

where C = correction constant

Pros: Handles uneven lighting
Cons: Slower computation

3. Niblack's Method

$$T(x, y) = \mu(x, y) + k \cdot \sigma(x, y)$$

k = sensitivity parameter (typically -0.2)
Best for: Noisy backgrounds

4. Sauvola's Method (Improved)

$$T(x, y) = \mu\left[1 + k\left(\frac{\sigma}{R} - 1\right)\right]$$

R = max std deviation, $k \in [0.2, 0.5]$
Best for: Document images with shadows

For OCR with Uneven Illumination

Recommended: Sauvola or Adaptive Gaussian — handles both shadows AND noise effectively.

Q7: OCR System — Binarization for Text Extraction

Question 7: OCR System with noisy, uneven-light scanned documents

(a) Explain digital image binarization and its importance in OCR.
(b) Describe methods; which handles uneven illumination and noise best?

(a) Binarization in OCR Context:

- Converts grayscale pixel values to 0 (background) or 255 (text)
- Isolates text characters from paper/shadow/stain
- Reduces data: 8-bit gray $\rightarrow$ 1-bit binary, reducing memory $8\times$
- Prerequisite for character recognition, contour extraction, feature detection

(b) Method Comparison for Noisy/Uneven Images:

Global (Otsu)

Fails when lighting varies — one threshold cannot separate all text regions.

Niblack

Works locally but **amplifies noise** in smooth regions due to std dev sensitivity.

Sauvola ✓Best

Locally adaptive + noise-resistant. Scales threshold with local contrast — **ideal for OCR**

Q8: Adaptive vs. Global Thresholding

Question 8: Binarization process; how adaptive thresholding improves over global

Global Thresholding (Otsu's)

1. Compute image histogram
2. Find T that maximizes σ_B^2
3. Apply: $B(x, y) = \mathbb{1}[f(x, y) \geq T]$

Example: $T = 128$ applied everywhere.
Problem: Left half in shadow $\rightarrow$ all background classified as text!

Failure Scenario

Image with bright region (gray $\approx$200) and dark region (gray $\approx$50). Global $T = 128$ misclassifies dark text on dark background.

Adaptive Thresholding

1. Divide image into $W \times W$ windows
2. For each window, compute μ_{local}
3. $T(x, y) = \mu_{local} - C$ (e.g., $C = 10$)
4. Apply local threshold per pixel

Result: Correct classification in both bright and dark regions!

Advantage

Illumination	Handled ✓
Shadow	Handled ✓
Noise	Reduced ✓

Part B

Image Segmentation Techniques

Clustering · K-Means · Fuzzy C-Means

Image Segmentation: Overview

Definition

Image segmentation partitions an image into **meaningful, non-overlapping regions** where pixels within a region share similar properties (color, intensity, texture).

Major Techniques:

1. **Thresholding**: Intensity-based
2. **Edge-based**: Detect boundaries (Canny, Sobel)
3. **Region-based**: Grow/split/merge regions
4. **Clustering**: Group similar pixels (K-means, FCM)
5. **Graph-based**: Min-cut, normalized cut
6. **Deep Learning**: U-Net, Mask R-CNN

Applications:

- Medical: Organ segmentation from MRI
- Autonomous driving: Road/pedestrian detection
- Satellite: Land use classification
- Industrial: Defect localization
- OCR: Character separation

Goal: Convert raw pixel data into structured, labeled regions for further processing.

Clustering-based Segmentation

Core Idea

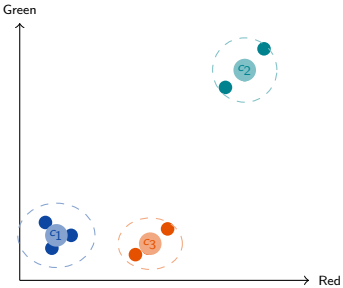
Cluster pixels into K groups based on feature similarity (intensity, color, texture). No labeled data needed — **unsupervised learning**.

Feature Space Representation:

- Grayscale: 1D feature — pixel intensity $[0, 255]$
- Color image: 3D feature — (R, G, B) or (L, a, b)
- Extended: (x, y, R, G, B) — spatial + color

Two Major Algorithms:

1. **K-Means**: Hard assignment — each pixel belongs to *exactly one* cluster
2. **Fuzzy C-Means**: Soft assignment — each pixel has *degree of membership* in each cluster



Pixel Feature Space ($K = 3$)

K-Means Clustering Algorithm

Algorithm Steps

1. Choose K (number of clusters)
2. **Initialize**: Randomly select K centroids $\{c_1, \ldots, c_K\}$
3. **Assignment**: Each pixel p_i assigned to nearest centroid:

$$\text{label}(p_i) = \arg\min_k \|p_i - c_k\|^2$$

4. **Update**: Recompute centroids:

$$c_k = \frac{1}{|S_k|} \sum_{p_i \in S_k} p_i$$

5. Repeat steps 3–4 until convergence

Objective Function

$$J = \sum^{K} \sum \|p_i - c_k\|^2 \quad \text{(minimizing)}$$

Simple Example (1D)

Pixel intensities: [10, 20, 80, 90, 150, 160]

Init: $c_1 = 10,\ c_2 = 80,\ c_3 = 150$

Assign:
10,20 $\rightarrow$ Cluster 1
80,90 $\rightarrow$ Cluster 2
150,160 $\rightarrow$ Cluster 3

Update:
$c_1 = (10 + 20)/2 = 15$
$c_2 = (80 + 90)/2 = 85$
$c_3 = (150 + 160)/2 = 155$

Converged! (No reassignment)

Segments: Dark / Mid / Bright

K-Means Clustering Algorithm

Algorithm Steps

1. Choose K (number of clusters)
2. **Initialize**: Randomly select K centroids $\{c_1, \ldots, c_K\}$
3. **Assignment**:
$$\text{label}(p_i) = \arg\min_k \|p_i - c_k\|^2$$
4. **Update**:
$$c_k = \frac{1}{|S_k|} \sum_{p_i \in S_k} p_i$$
5. Repeat steps 3–4 until convergence

Objective Function

$$J = \sum_{k=1}^{K} \sum_{p_i \in S_k} \|p_i - c_k\|^2$$

K-Means Clustering: Example (1D)

Simple Example

Pixel intensities: [10, 20, 80, 90, 150, 160]

Init: $c_1 = 10,\ c_2 = 80,\ c_3 = 150$

Assign:
10,20 $\rightarrow$ Cluster 1
80,90 $\rightarrow$ Cluster 2
150,160 $\rightarrow$ Cluster 3

Update:
$c_1 = 15,\ c_2 = 85,\ c_3 = 155$

Converged! (No reassignment)

Segments: Dark / Mid / Bright

Refer previous slide for algorithm steps

Q9: K-Means for Image Segmentation — Detailed Example

Question 9: K-Means clustering for image segmentation with example

Image: 8 pixels with intensities: $\{45, 48, 50, 100, 105, 110, 200, 205\}$, $K = 3$

Iteration 1:

Init centroids: $c_1 = 45,\ c_2 = 100,\ c_3 = 200$

Distances for pixel 50:
$|50 - 45| = 5,\ |50 - 100| = 50,\ |50 - 200| = 150$
$\Rightarrow$ Assigned to c_1

After assignment:
Cluster 1: $\{45, 48, 50\}$, new $c_1 = 47.7$
Cluster 2: $\{100, 105, 110\}$, new $c_2 = 105$
Cluster 3: $\{200, 205\}$, new $c_3 = 202.5$

Iteration 2: Reassign with updated centroids

No pixel changes cluster $\Rightarrow$ **Converged!**

Cluster	Pixels	Label
1	45, 48, 50	Dark (shadow)
2	100, 105, 110	Mid-tone
3	200, 205	Bright (text?)

Segment boundary: between 50 and 100

Fuzzy C-Means (FCM) Algorithm

Core Difference from K-Means

Each pixel p_i has a **membership degree** $u_{ik} \in [0, 1]$ for each cluster k, with:

$$\sum_{k=1}^{C} u_{ik} = 1 \quad \forall i$$

FCM Objective Function:

$$J_m = \sum_{i=1}^{N} \sum_{k=1}^{C} u_{ik}^m \|p_i - c_k\|^2$$

where $m > 1$ is fuzziness parameter (typically $m = 2$)

Update Rules:

$$u_{ik} = \frac{1}{\sum_{c} \quad ^2}$$

FCM Example

Pixel intensity $= 130$
Cluster centers: $c_1 = 50$ (dark), $c_2 = 200$ (bright)

Distances: $|130 - 50| = 80$, $|130 - 200| = 70$

With $m = 2$:

$$u_1 \approx \frac{1}{1 + (80/70)^2} \approx 0.43$$

$$u_2 \approx \frac{1}{1 + (70/80)^2} \approx 0.57$$

Result: 43% dark, 57% bright

$\Rightarrow$ Pixel is on the boundary!

Fuzzy C-Means (FCM) Algorithm

Core Difference from K-Means

Each pixel p_i has a **membership degree** $u_{ik} \in [0, 1]$ for each cluster k, with:

$$\sum_{k=1}^{C} u_{ik} = 1 \quad \forall i$$

FCM Objective Function:

$$J_m = \sum_{i=1}^{N} \sum_{k=1}^{C} u_{ik}^m \|p_i - c_k\|^2$$

where $m > 1$ is the fuzziness parameter (typically $m = 2$)

Update Rules:

$$u_{ik} = \frac{1}{\sum_{j=1}^{C} \left(\frac{\|p_i - c_k\|}{\|p_i - c_j\|} \right)^{\frac{2}{m-1}}}$$

$$c_k = \frac{\sum_{i=1}^{N} u_{ik}^m \, p_i}{\sum_{i=1}^{N} u_{ik}^m}$$

Fuzzy C-Means: Example

FCM Example

Pixel intensity $= 130$

Cluster centers: $c_1 = 50$ (dark), $c_2 = 200$ (bright)

Distances: $|130 - 50| = 80$, $|130 - 200| = 70$

With $m = 2$:

$$u_1 \approx \frac{1}{1 + (80/70)^2} \approx 0.43$$

$$u_2 \approx \frac{1}{1 + (70/80)^2} \approx 0.57$$

Result: 43% dark, 57% bright

Key Insight

Boundary pixels get *partial* membership — more realistic than hard assignment.

Refer previous slide for equations

Q10: K-Means vs. Fuzzy C-Means Comparison

Question 10: Compare K-means and Fuzzy C-means for image segmentation

Property	K-Means
Membership	Hard (0 or 1)
Assignment	Pixel $\in$ exactly 1 cluster
Boundary pixels	Forced to nearest cluster
Objective	Sum of squared distances
Uncertainty	Not handled
Parameter	K only
Convergence	Fast
Noise sensitivity	High
Best for	Well-separated regions

Property	Fuzzy C-Means
Membership	Soft ($u_{ik} \in [0, 1]$)
Assignment	Partial across all clusters
Boundary pixels	Gets partial memberships
Objective	Weighted sum with fuzziness
Uncertainty	Explicitly modeled
Parameter	C and fuzziness m
Convergence	Slower
Noise sensitivity	Lower
Best for	Overlapping/ambiguous regions

Q11: FCM — Handling Uncertainty vs. K-Means

Question 11: Working principle of FCM; uncertainty handling vs K-means

FCM Working Principle:

1. Initialize membership matrix U randomly
2. Compute fuzzy cluster centroids using u_{ik}^m
3. Update memberships based on distances to all centroids
4. Repeat until $\|U^{new} - U^{old}\| < \epsilon$

Handling Uncertainty:

- **Boundary pixels**: Both u_{ik} values ≈ 0.5 — algorithm "knows" it's uncertain
- **Clear pixels**: One $u_{ik} \approx 1$, others ≈ 0
- **Noisy pixels**: Spread membership prevents hard

Medical Imaging Case

In brain MRI segmentation:

- White matter, gray matter, CSF regions overlap
- K-Means forces each voxel to one tissue type
- FCM gives partial memberships, enabling probabilistic tissue maps
- Result: Clinically more accurate volume estimates

Fuzziness Parameter m

Q13: Clustering-based vs. Edge-based Segmentation

Question 13: Compare clustering-based and edge-based segmentation; advantages & limitations

Clustering-based

Method: Groups pixels by feature similarity without detecting boundaries explicitly.

Advantages:

- Unsupervised — no labels needed
- Handles textures and gradual transitions
- Works well with color/multi-spectral
- Robust to weak edges

Limitations:

- Must specify K in advance
- Sensitive to initialization

Edge-based (Canny, Sobel, etc.)

Method: Detects pixel intensity discontinuities as boundaries between regions.

Advantages:

- Preserves sharp, fine boundaries
- Fast computation
- Intuitive — matches visual boundaries
- Works well with simple scenes

Limitations:

- Open contours (gaps in edges)
- Very sensitive to noise

Part C

Fuzzy Logic, Neural Networks & Color Images

Object Extraction · Color Challenges · Deep Learning

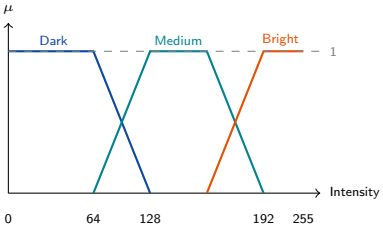
Fuzzy Logic in Digital Image Processing

Why Fuzzy Logic in DIP?

Digital images contain inherent **vagueness and uncertainty**:

- "Bright" vs. "dark" — no sharp boundary
- Edge vs. non-edge pixels on gradients
- Noisy pixels may belong to multiple regions

Fuzzy logic handles this with **membership functions** $\mu(x) \in [0, 1]$.



Fuzzy Membership Functions

Applications:

- **Image Enhancement**: Fuzzy histogram equalization
- **Edge Detection**: Fuzzy Sobel, fuzzy gradient operators
- **Noise Filtering**: Fuzzy median filters
- **Segmentation**: FCM (as discussed)

Fuzzy IF-THEN Rule

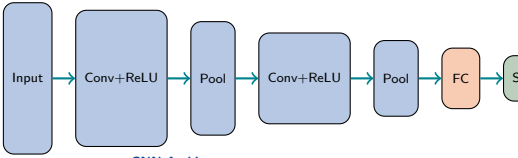
IF pixel is *Dark AND* neighbor is *Medium THEN* enhance to *Medium*

Neural Networks in Digital Image Processing

Convolutional Neural Networks (CNNs)

CNNs learn hierarchical features automatically:

1. **Conv layers**: Extract local features (edges, textures, shapes)
2. **Pooling**: Spatial downsampling
3. **Fully connected**: High-level classification



CNN Architecture

Key Architectures for DIP:

- **U-Net**: Encoder-decoder with skip connections (medical segmentation)
- **Mask R-CNN**: Instance segmentation
- **DeepLab**: Atrous convolutions for semantic seg.
- **GAN**: Image-to-image translation, enhancement
- **ResNet/VGG**: Feature extraction backbone

Advantages over Traditional

- Learns features automatically
- End-to-end trainable
- Handles complex backgrounds
- State-of-the-art accuracy

Problems with Color Images

Major Challenges

1. **High Dimensionality**: 3 channels (RGB) → clustering in 3D space; more complex than grayscale
2. **Color Space Selection**: RGB is not perceptually uniform; HSV, Lab better for segmentation
3. **Illumination Variation**: Same object looks different under different lighting
4. **Shadow and Highlights**: Alter perceived color drastically
5. **Color Similarity**: Different objects may have same color
6. **Color Bleeding**: Adjacent regions bleed colors at boundaries
7. **Compression Artifacts**: JPEG compression introduces chroma noise

Color Space Comparison:

Space	Properties
RGB	Device-dependent; not uniform
HSV	Separates hue from intensity
Lab	Perceptually uniform; illumination-invariant
YCbCr	Luminance-chrominance separation

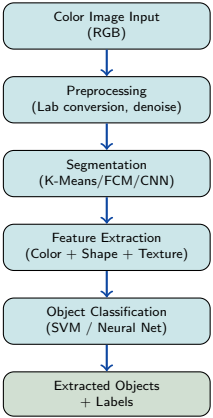
Best Practice

Convert RGB → **CIE Lab** before segmentation for more stable clustering results

Object Extraction from Color Images

Pipeline for Object Extraction:

1. **Preprocessing**: Denoise, normalize, color space conversion
2. **Segmentation**: K-means/FCM/DL to find regions
3. **Feature Extraction**: Color histograms, SIFT, HOG, CNN features
4. **Object Localization**: Bounding boxes, region proposals
5. **Classification/Labeling**: SVM, CNN, or fuzzy rules
6. **Post-processing**: Morphological ops, boundary refinement



Neural Network Approaches

- **Faster R-CNN**: Region proposals + CNN classification
- **YOLO**: Single-pass detection, real-time

Q12: Challenges in Color Object Extraction + Neural Networks

Question 12: Challenges in extracting objects from color images; neural network improvements

Challenges:

- **Color ambiguity**: Multiple objects share similar hues
- **Illumination**: Shadows change apparent color
- **Occlusion**: Objects partially hidden by others
- **Scale variation**: Objects at different distances
- **Background clutter**: Complex textures compete
- **Deformable shapes**: Objects change appearance
- **Class imbalance**: Rare object categories

Neural Network Solutions:

- **Data augmentation**: Teaches invariance to color shift, scale, rotation
- **Attention mechanisms**: Focus on relevant color regions
- **Multi-scale detection**: Feature Pyramid Networks (FPN)
- **Transfer learning**: Pre-trained features from ImageNet
- **Instance segmentation**: Mask R-CNN separates touching objects
- **Color normalization layers**: Network

Q14: Fuzzy Logic & Neural Networks in DIP — Role, Applications, Limitations

Question 14: Categorize roles of fuzzy logic and neural networks; applications and limitations

Fuzzy Logic

Role: Handle vagueness in pixel classification, rule-based enhancement

Applications:

- Adaptive image enhancement
- Edge detection under noise
- Medical image analysis (MRI tissues)
- Fuzzy morphology operations

Limitations:

- Membership functions need expert design

Neural Networks

Role: Learn complex patterns from data; replace hand-crafted features

Applications:

- Semantic/instance segmentation
- Object detection (YOLO, R-CNN)
- Super-resolution, denoising
- Medical diagnosis from images

Limitations:

- Requires large labeled datasets

Q15: Neural Networks in Segmentation & Color Image Challenges

Question 15: Neural networks in image segmentation; challenges in handling color images

Neural Networks for Segmentation:

U-Net (Encoder-Decoder):

- Encoder: Downsamples, extracts features
- Decoder: Upsamples, reconstructs segmentation map
- Skip connections: Preserve spatial details
- Excellent for small datasets (medical imaging)

DeepLabV3+:

- Atrous convolutions for multi-scale context
- ASPP module captures global information
- State-of-the-art on Pascal VOC, Cityscapes

Color Image Challenges for NNs:

- Training on RGB vs. Lab — model must learn color normalization
- $3\times$ more parameters than grayscale
- Color bias in training data affects generalization
- Domain shift: indoor training $\rightarrow$ outdoor failure

Solutions

- Batch normalization per channel
- Color jitter augmentation

Q16: Four Major Application Areas of DIP

Question 16: List four major application areas; describe one in detail

Four Major Areas:

1. **Medical Imaging**
2. **Remote Sensing & Satellite Analysis**
3. **Industrial Inspection & Quality Control**
4. **Biometric Security (Face/Fingerprint)**

Detailed: Medical Imaging

Goal: Extract clinically relevant information from medical scans (X-ray, MRI, CT, Ultrasound).

Techniques Used:

- Enhancement: CLAHE, histogram equalization

Medical Imaging Applications

Domain	DIP Task
Radiology	Lung nodule detection in CT
Pathology	Cell counting in slides
Ophthalmology	Retinal vessel segmentation
Cardiology	Heart wall motion analysis
Neurology	Brain tumor boundary detection

Unit 2 Summary

Part A: Foundations

- Digital image = pixel matrix $f(x, y)$
- DIP applied in medicine, security, industry, remote sensing
- Binarization: grayscale $\rightarrow$ 0/1 using threshold T
- Otsu: global; Sauvola: adaptive (best for OCR)

Part B: Segmentation

- Segmentation partitions image into meaningful regions
- K-Means: hard assignment, minimizes $\sum \|p_i - c_k\|^2$
- FCM: soft membership $u_{ik} \in [0, 1]$, handles uncertainty

Part C: Advanced

- Fuzzy logic: models vagueness via membership functions
- Neural networks: learn features automatically from data
- Color images: 3D feature space, illumination-sensitive
- Lab color space preferred over RGB for segmentation
- Deep learning (Mask R-CNN, U-Net) best for object extraction

Exam Tips (Q7–Q16)

- Always state the objective function for K-means and FCM
- For binarization: mention Otsu vs. Sauvola

Thank You!

Unit 2: Digital Image Processing

Key Topics Covered:

Binarization · K-Means · Fuzzy C-Means · Fuzzy Logic · Neural Networks · Color Images

Questions 7 – 16 fully solved

Department of Computer Science